

Title:
MOBILE PHONE PRICE PREDICTION
USING MACHINE LEARNING

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ABSTRACT

The rapid growth of the smartphone industry has led to a wide variety of mobile devices with different specifications and price ranges. Predicting the price category of a mobile phone based on its features is a challenging task that can be effectively addressed using machine learning techniques. This project focuses on developing a machine learning-based classification system to predict the price range of mobile phones using their technical specifications.

The dataset used in this project includes features such as battery power, RAM, camera quality, screen resolution, and connectivity options. These features are analyzed to identify patterns that influence the price category of mobile devices. The target variable classifies phones into four categories: low cost, medium cost, high cost, and very high cost.

Various machine learning algorithms such as Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, and Support Vector Machine are implemented and compared. The Random Forest model achieved the best performance and was selected as the final model.

The results demonstrate that machine learning can effectively predict mobile price categories, which can be useful for both consumers and manufacturers in decision-making.

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INTRODUCTION

With the increasing demand for smartphones, there is a wide variety of mobile phones available in the market with different specifications and price ranges. Consumers often find it difficult to choose a mobile phone that fits their budget and requirements. Therefore, predicting the price category of a mobile phone based on its specifications is an important task.

Machine learning provides an effective way to analyze large datasets and identify patterns that influence pricing. By using features such as RAM, battery capacity, camera quality, and connectivity options, machine learning models can classify mobile phones into different price categories.

In this project, a classification model is developed to predict the price range of mobile phones. The model categorizes phones into four classes: low cost, medium cost, high cost, and very high cost.

The main objective of this project is to build an accurate and efficient prediction model that can assist users and manufacturers in understanding pricing trends. The report explains the entire workflow, including data preprocessing, model training, and evaluation.

PROBLEM STATEMENT

The mobile phone market is highly competitive, with numerous devices offering different specifications at varying price ranges. Determining the appropriate price category of a mobile phone based on its features is a complex problem.

Traditional methods rely on manual analysis and market research, which can be time-consuming and inefficient. Therefore, there is a need for an automated system that can predict the price category based on technical specifications.

The problem addressed in this project is to develop a machine learning model that can classify mobile phones into different price ranges using their features. The model should accurately distinguish between multiple classes and provide reliable predictions.

Challenges include handling multiple features, selecting the most relevant attributes, and choosing the best algorithm for multiclass classification.

OBJECTIVES OF THE PROJECT

The main objective of this project is to develop a machine learning model that can accurately predict the price range of mobile phones.

Other objectives include analyzing the dataset to understand the importance of different features, performing data preprocessing to improve data quality, and implementing multiple machine learning algorithms for comparison.

The project also aims to evaluate model performance using metrics such as accuracy and precision, and to select the best-performing model for final prediction.

LITERATURE REVIEW

Mobile price prediction has been widely studied using machine learning techniques. Traditional pricing methods relied on manual analysis, but recent approaches use data-driven models.

Various algorithms such as Decision Trees, Random Forest, and Support Vector Machines have been successfully applied for classification tasks. Ensemble methods like Random Forest have shown higher accuracy due to their ability to handle complex data.

Feature selection and preprocessing play a significant role in improving model performance. Studies suggest that selecting relevant features leads to better accuracy.

Overall, machine learning has proven to be an effective tool for price prediction problems.

DATASET DESCRIPTION

The dataset used in this project contains various features related to mobile phone specifications. These include battery power, RAM, camera quality, screen size, and connectivity features such as Bluetooth, WiFi, and 4G support.

The target variable is the price range, which is divided into four categories: low cost, medium cost, high cost, and very high cost.

The dataset is analyzed to understand feature distribution and relationships. It is then divided into training and testing sets for model development and evaluation.

METHODOLOGY

The project follows a systematic approach starting with data collection and preprocessing. Exploratory Data Analysis (EDA) is performed to understand patterns and relationships in the dataset.

The dataset is then split into training and testing sets. Multiple machine learning models are trained and evaluated using performance metrics.

The best-performing model is selected based on accuracy and used for prediction.

DATA PREPROCESSING

Data preprocessing involves cleaning the dataset and preparing it for model training. This includes handling missing values, scaling features, and encoding categorical variables.

Feature selection is performed to remove irrelevant attributes and improve model performance. Normalization ensures that all features contribute equally.

Proper preprocessing improves the accuracy and efficiency of the model.

MODEL IMPLEMENTATION

Several machine learning models are implemented, including Logistic Regression, KNN, Decision Tree, Random Forest, and SVM.

Each model is trained using the dataset and evaluated based on performance metrics. Random Forest achieved the best results due to its ensemble learning capability.

Hyperparameter tuning is performed to optimize model performance.

RESULTS AND EVALUATION

The model achieved an overall accuracy of **89%**, indicating good performance in classifying mobile phones into the correct price ranges.

The major evaluation metrics are:

- **Accuracy:** 89%
- **Precision (average):** 0.89
- **Recall (average):** 0.89
- **F1-Score (average):** 0.89

Class-wise performance:

- **Class 0 (Low Cost):** Precision = 0.95, Recall = 0.96, F1 = 0.96
- **Class 1 (Medium Cost):** Precision = 0.89, Recall = 0.87, F1 = 0.88
- **Class 2 (High Cost):** Precision = 0.78, Recall = 0.87, F1 = 0.82
- **Class 3 (Very High Cost):** Precision = 0.94, Recall = 0.87, F1 = 0.90

Overall, the model performs well across all classes, with slightly lower precision for the high-cost category.

CONCLUSION

This project demonstrates the effectiveness of machine learning in predicting mobile phone price ranges. The model successfully classifies phones into different categories based on their features.

Random Forest proved to be the best-performing model. The system can assist consumers and manufacturers in decision-making.

FUTURE SCOPE

The project can be improved by using deep learning techniques and larger datasets. Real-time prediction systems can also be developed.

Integration with e-commerce platforms can enhance usability.

VISUALIZATION

Visualization techniques such as bar charts, histograms, and heatmaps were used to analyze the dataset. These plots helped in understanding feature distribution and relationships.

Correlation heatmaps were used to identify important features. Confusion matrix was used to evaluate model performance.

Visualization improved data understanding and model evaluation.

CHALLENGES FACED

The project faced challenges such as handling multiple features, selecting the best model, and tuning hyperparameters.

Multiclass classification added complexity compared to binary classification. Ensuring model accuracy across all classes was difficult.

Despite these challenges, the project was successfully completed.

REFERENCES

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